An Interactive Genetic Algorithm Approach for Generating Aesthetic Ita Bag Layouts

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Agenda:

- Introduction & Background
- System Overview
- Dataset
- Experiment
- Conclusion

Introduction & Background: – What is Ita Bag

- **What is an Ita Bag**
  - A type of bag decorated with anime/game merchandise (badges, pins, charms).
  - Popular in fan cultures, especially in Japan

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Introduction & Background: – Example of Ita Bag

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Introduction & Background: – Problem

- **The Problem:**
  - – Layout is **manual**, **trial-and-error**, and often **frustrating**
  - – No existing **intelligent tools** to help users optimize design

Related Work & Project Goal: – Existing Tools

- **Existing Methods:**
  - Manual design only; users rely on personal intuition
- **Some layout optimization in:**
  - **UI/UX design tools**
  - **Advertisement placement systems**
  - **Packing & tiling algorithms**

Related Work & Project Goal: – Existing Tools

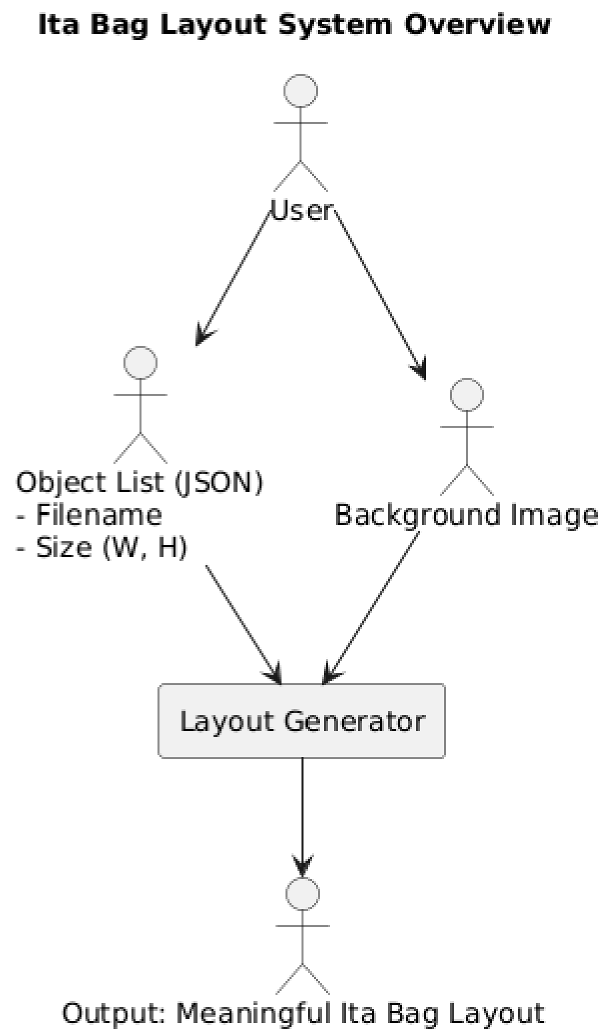
- **Limitations:**
  - No method considers **aesthetic preferences**
  - Existing methods lack **user interaction**
  - Not adaptable to **individual tastes**

Related Work & Project Goal

- **Project Goal:**
  - Develop a layout generation system
  - Support user feedback to guide layout optimization
  - Adapt to individual aesthetic preferences

System Overview: – Proposed

- **Input:**
  - List of items
  - Background image
- **Output:**
  - Clean, well-arranged layout



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Prototype I

- **Prototype**
  - Multi-Layer Perceptron (MLP) model trained on 200 samples
  - Promising results, but lacked global awareness
- **Limitations**
  - Needs ≥ 1,000 samples
  - Struggles with overall layout composition

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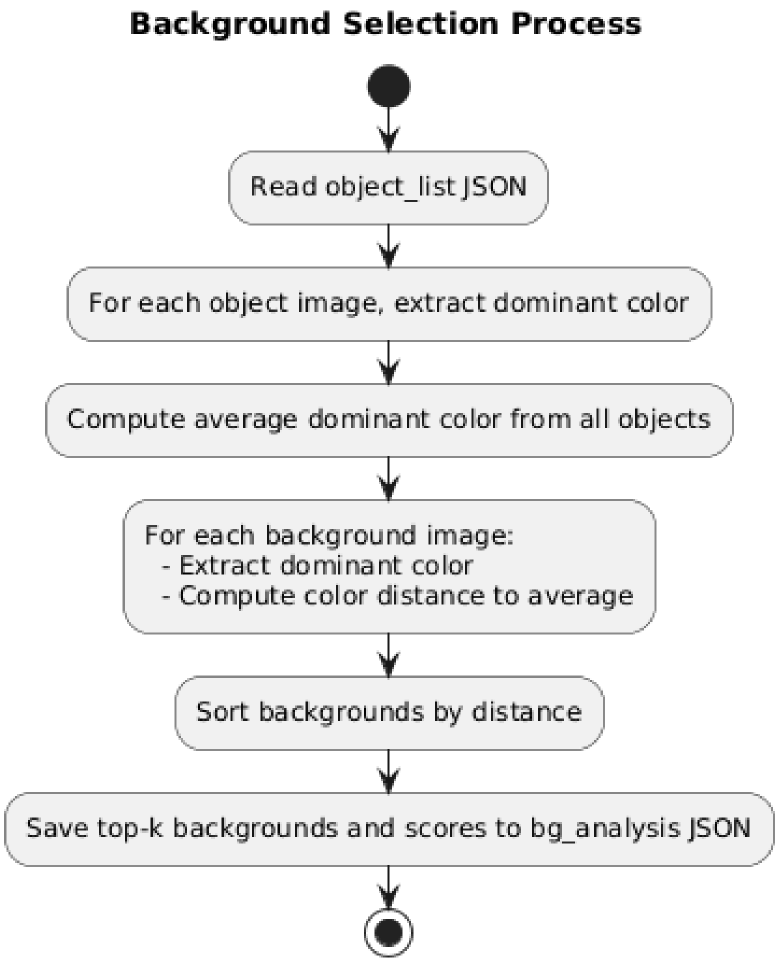
Prototype II

- **Design Shift**
  - Moved to Interactive Genetic Algorithm (IGA)
  - Receive direct feedback from users

Dataset:
– Items & Backgrounds

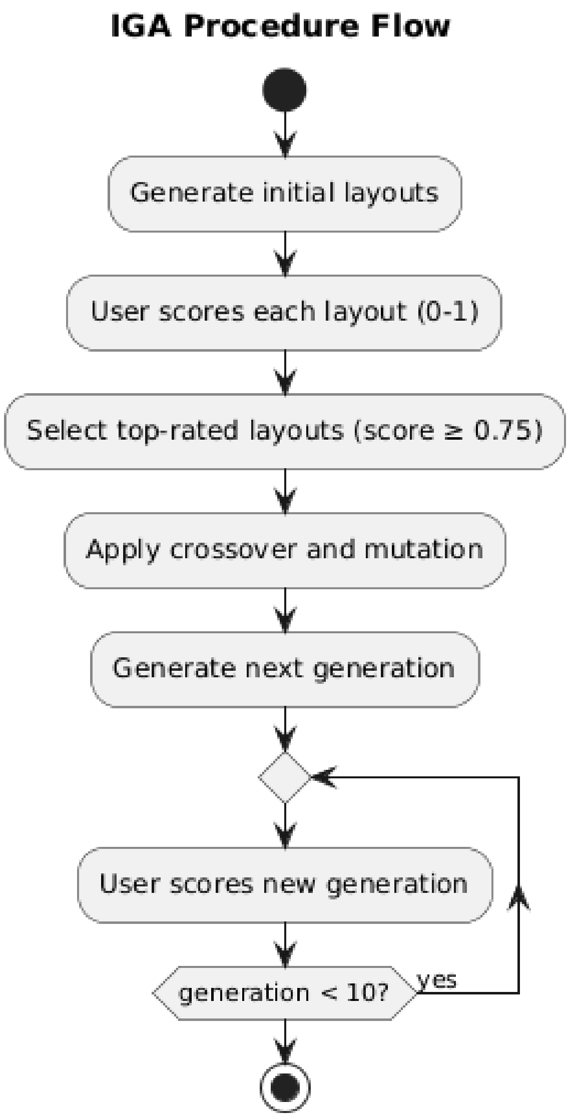
- **Object Dataset:**
  – object_list_00X.json
    - Filename
    - size (width, height)
- **Background Images:**
  – Background assets with diverse colors, textures, and styles

Dataset: – Background Selection



1. Extract dominant
2. Compute color distance
3. Select backgrounds

Experiment & Setup



- 8 layouts per generation
- User gives a score from 0 to 1
- Best layouts selected (score 0.75)
- Apply crossover & mutation
- Repeat for 10 generations
- Save outputs & plot trend

Scoring Criteria

- **Reference Scoring Criteria**
  - **Proximity:** Measures how closely related elements are placed together
  - **White Space:** Evaluates the amount and distribution of empty space
  - **Alignment:** Assesses how well elements line up horizontally or vertically

Limitations

- **User scores remained flat across generations**
  - Little difference observed between Generation 1 and Generation 10
- **Possible reasons:**
  - Subjective scoring
  - Dominant object constraint
  - Reference vs. user scores

Result:
– With Dominant Object

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Result:
– Without Dominant Object

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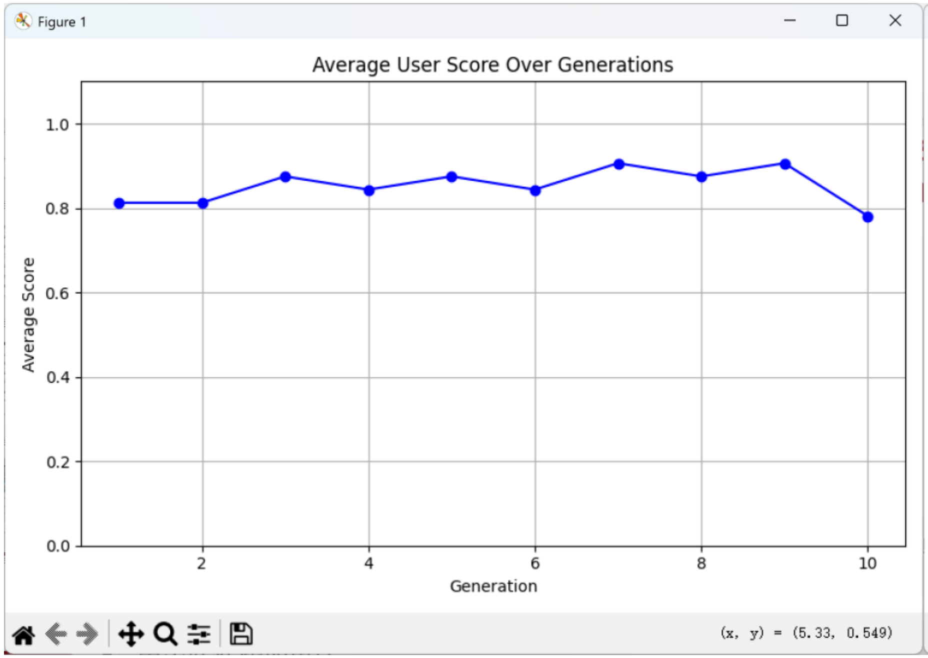
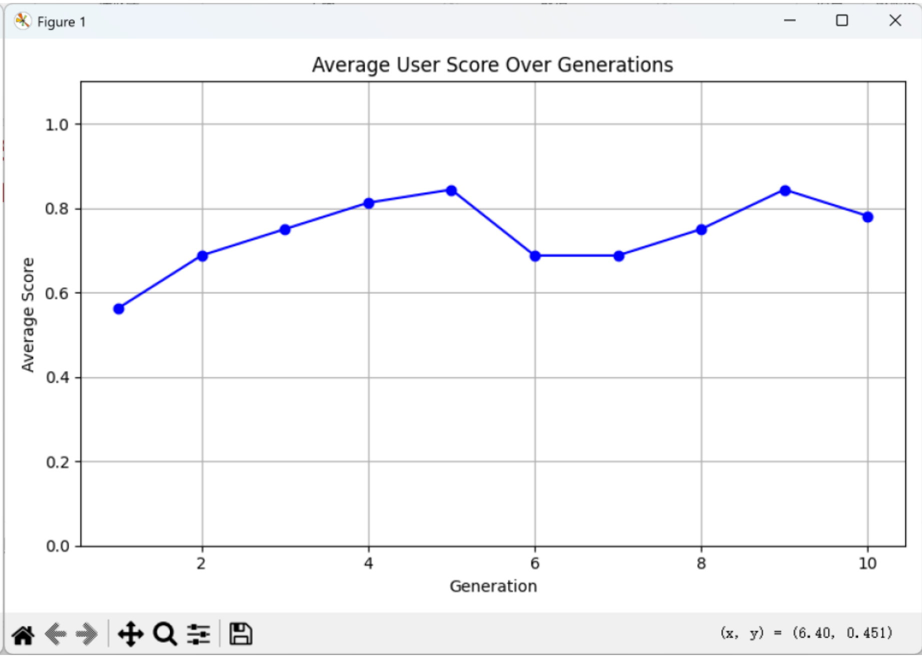
Observation

- Result with dominant & without dominant object





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Discussion: – Impact of a Dominant

- **With dominant object**
  - higher average scores
  - layouts feel more structured
- **Without dominant object**
  - lower average scores
  - layouts feel random or messy

Conclusion

- **Conclusion**
  - Dominant objects improve visual focus and scores but limited layout diversity and weak generational change.
- **Future Work**
  - Improve scoring methods

Q & A